

Locally Compact Groups

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Topological Primer

Definition. A **topological group** G is a group G equipped with the topology such $(x, y) \rightarrow xy$ and $i : x \rightarrow x^{-1}$ are continuous. $\forall g \in G, L_g : G \rightarrow G$ is left multiplication by g . ■

Definition. A **symmetric** neighborhood V is a neighborhood such that $V = V^{-1}$. ■

Claim 2.1. *Let G be a topological group:*

1. *The topology of G is invariant under translations and inversions, that is $\forall U \in \mathcal{T}_G, \forall g \in G$
 $gU, Ug, U^{-1} \in \mathcal{T}_G$. Furthermore $\forall A \subseteq G, AU, UA \in \mathcal{T}_G$*
2. *$\forall U \in \mathcal{T} \ni 1 \in U, \exists V$ symmetric such that $VV \subseteq U$*
3. *If $H \leq G \implies \overline{H} \leq G$*
4. *Every open subgroup of G is closed*
5. *If A and B are compact sets in G , so is AB*

Proof. 1. $L_{g^{-1}}^*(U) = gU, L_g^*(U) = g^{-1}U, i^*(U) = U^{-1}$ are all open by continuity. $AU = \bigcup_{a \in A} aU$ and $UA = \bigcup_{a \in A} Ua$ are open by closure of unions in topology.

2. Continuity of $(x, y) \rightarrow xy$ means that $\exists W_1, W_2 \in \mathcal{T} \ni 1 \in W_1 \cap W_2, W_1 W_2 \subseteq U$. Take $V = W_1 \cap W_2 \cap W_1^{-1} \cap W_2^{-1}$.

3. Just prove inverses and closure of multiplication exist. $\forall x \in \overline{H} \setminus H, \exists U_x \in \mathcal{T} \ni x \in H \cap U_x$

□

Definition. G is a **locally compact group** if and only if G is hausdorff and locally compact (every point has a compact neighborhood). ■

Haar Measure

Definition. A **left Haar Measure** on G is a nonzero Radon measure (???) μ on G such that $\mu(xE) = \mu(E)$ for every Borel set $E \subseteq G$ and $\forall x \in G$ ■

Claim 2.9. Let μ be a Radon measure on a locally compact group G , and let $\tilde{\mu}(E) = \mu(E^{-1})$

1. μ is a left Haar measure $\iff \tilde{\mu}$ is a right Haar measure
2. μ is a left Haar measure if and only if $\int L_y f d\mu = \int f d\mu$ for every $f \in C_c^+(G)$ and every $\forall y \in G$.

Proof. 1. $\implies$ Suppose μ is a left Haar measure. $\forall E \subseteq G$, borel, and $\forall x \in G$, $\tilde{\mu}(Ex) = \mu((Ex)^{-1}) = \mu(x^{-1}E^{-1}) = \mu(E^{-1}) = \tilde{\mu}(E^{-1})$. $\Leftarrow$: Reverse the last thing.

2. $\implies$: Suppose μ is a Haar measure, then $\int L_y f d\mu = \int f d\mu_y = \int f d\mu$ for all f , by approximating f by simple functions. $\Leftarrow$: Suppose $\forall f \in C_c^+(G)$, $\int L_y f d\mu = \int f d\mu_y = \int f d\mu$, by uniqueness of Riesz representation $\mu = \mu_y$.

□

Claim 2.10. Every locally compact group has a left Haar measure λ

Claim 2.21. Suppose the underlying structure of G is a open subset of $\mathbb{R}^n$ and left translations are given by $xy = A(x)y + b(x)$ for $A(x) \in Gl_n(\mathbb{R})$ and $y \in \mathbb{R}^n$. Then $|\det(A(x))|^{-1} d\mu$ where $d\mu$ is the standard Lebesgue measure on $\mathbb{R}^n$.

Proof.

□